

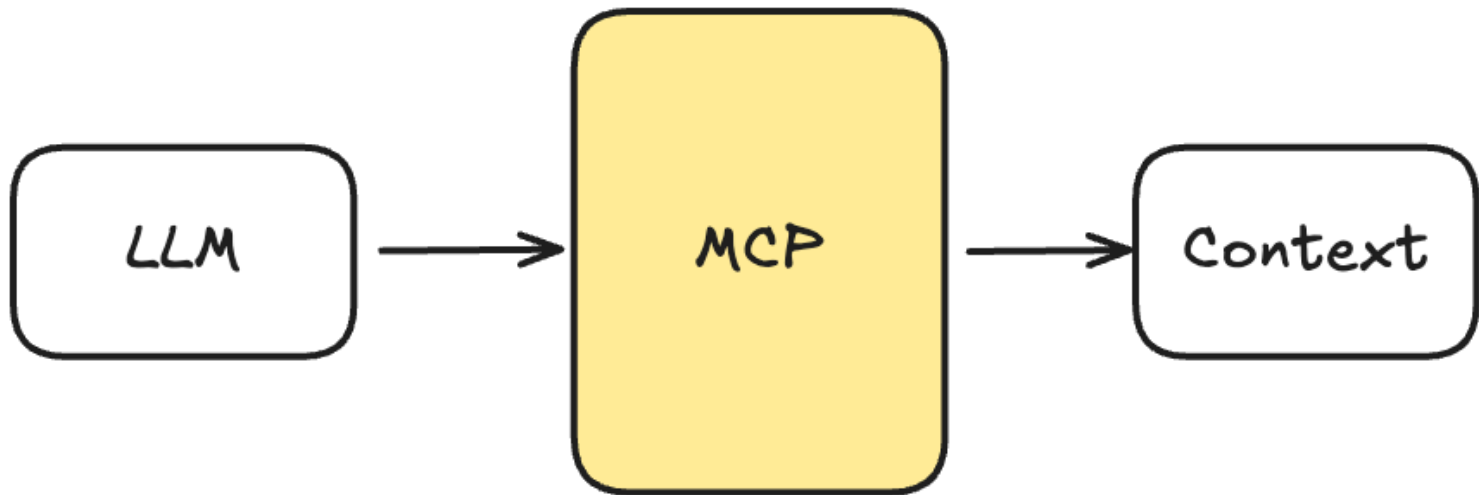
Building Agents with MCP

The HTTP Moment of AI?

Introduction to MCP

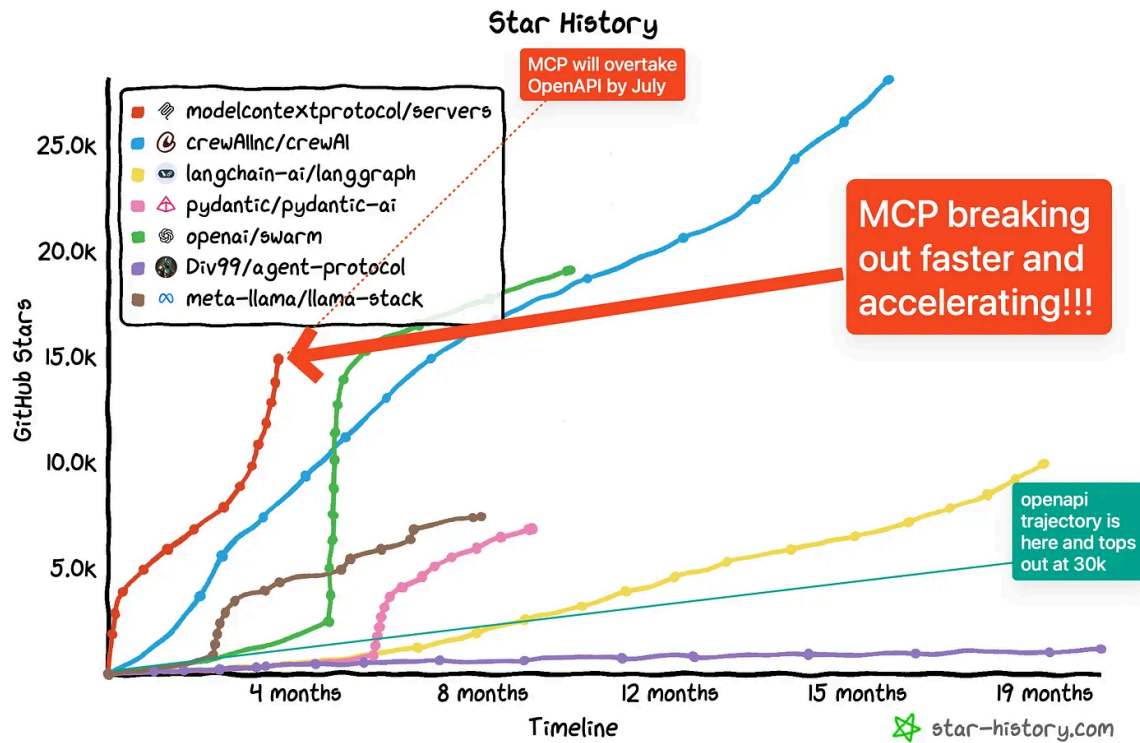
What is MCP?

Open Protocol to standardize connections between LLMs and Context

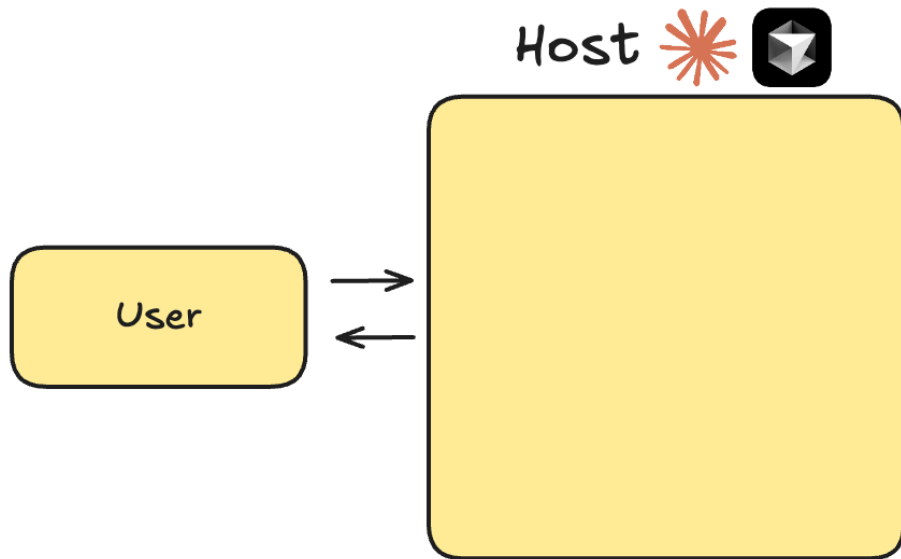


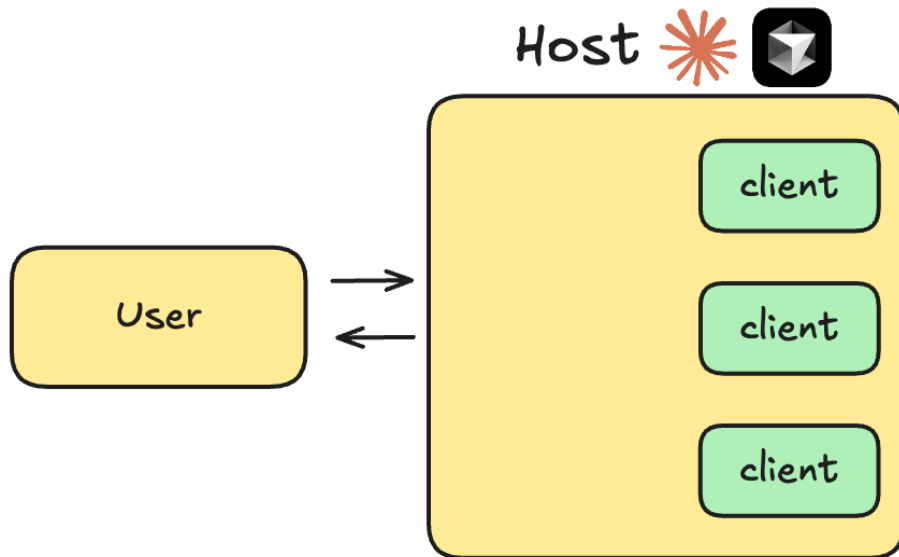
MCP is what makes AI actually useful for real apps

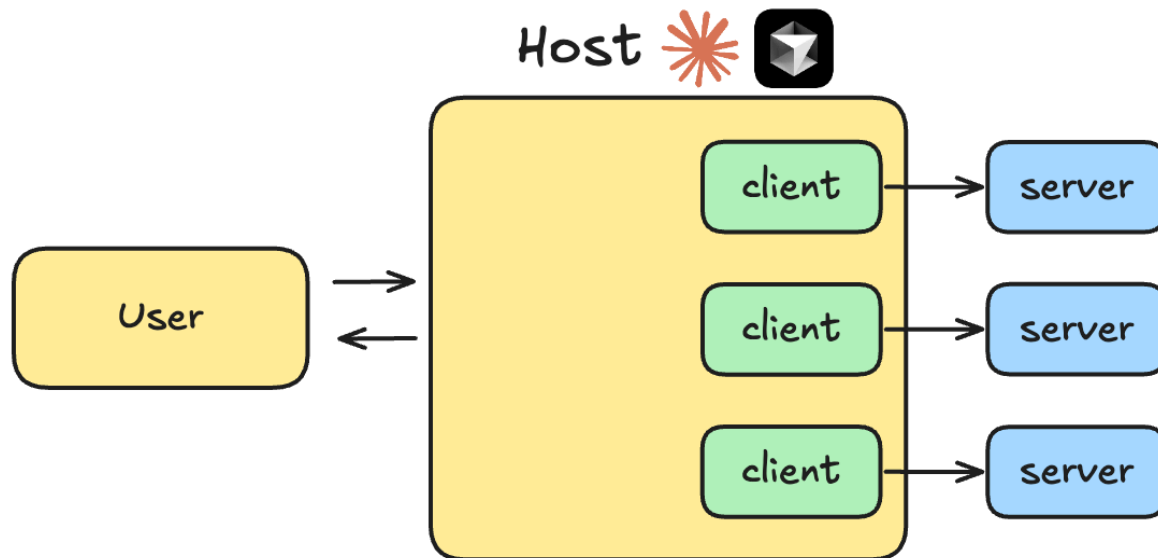
MCP is Growing Super Fast

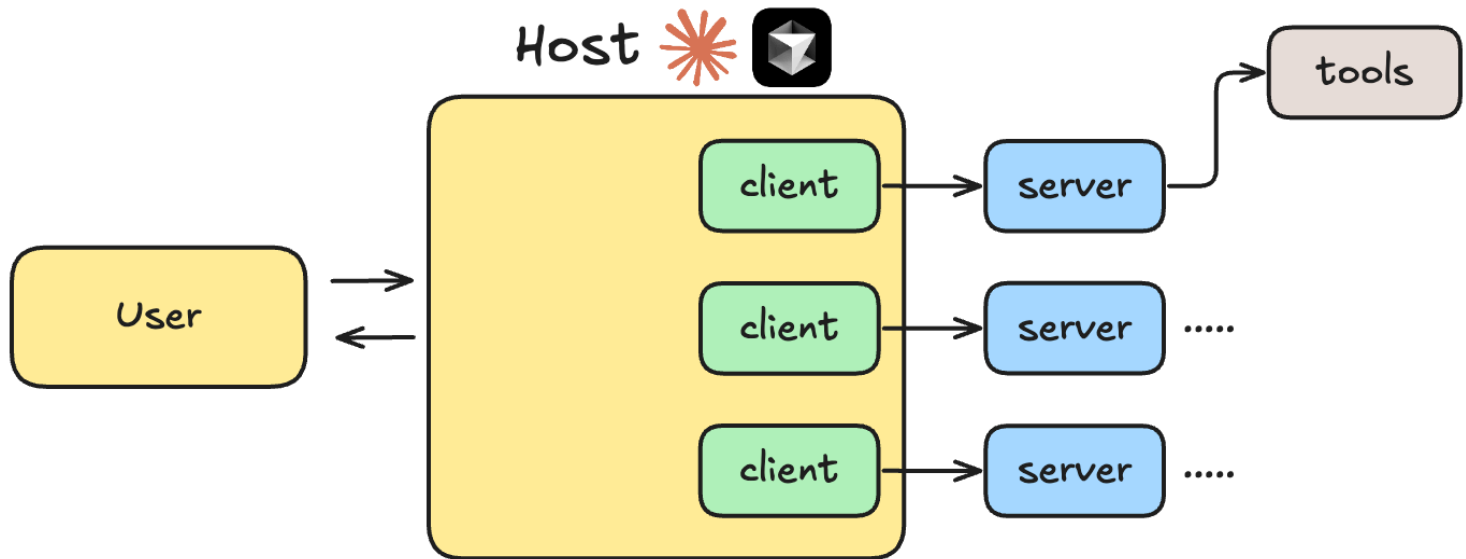


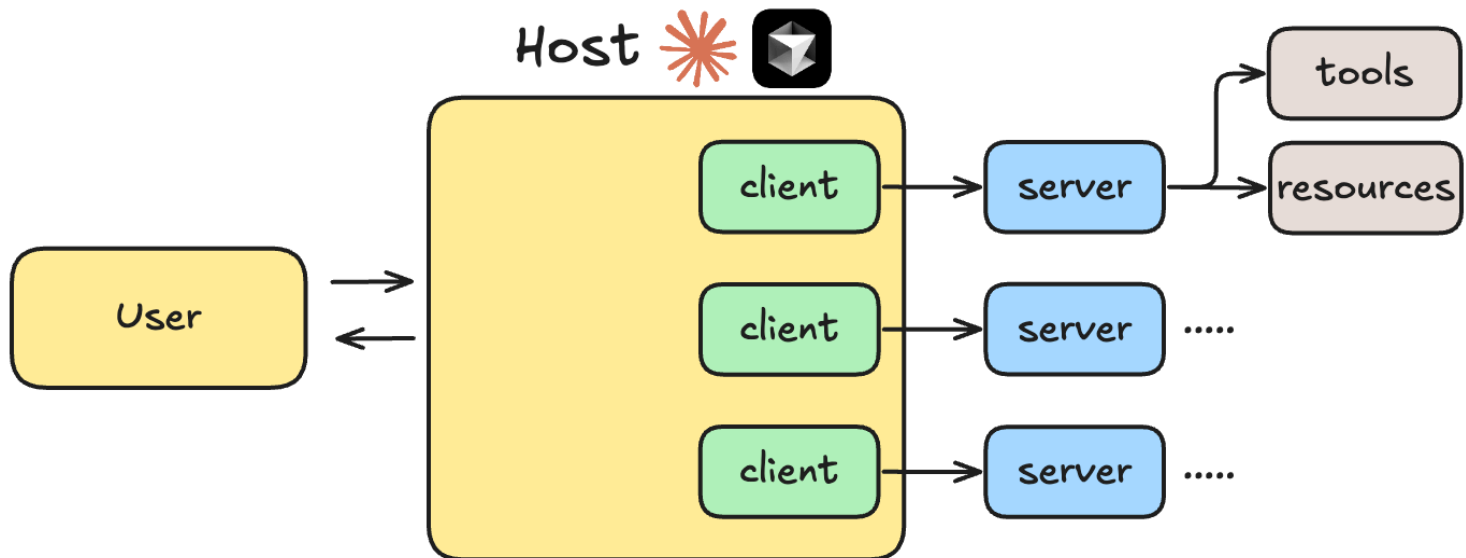
MCP Core Components

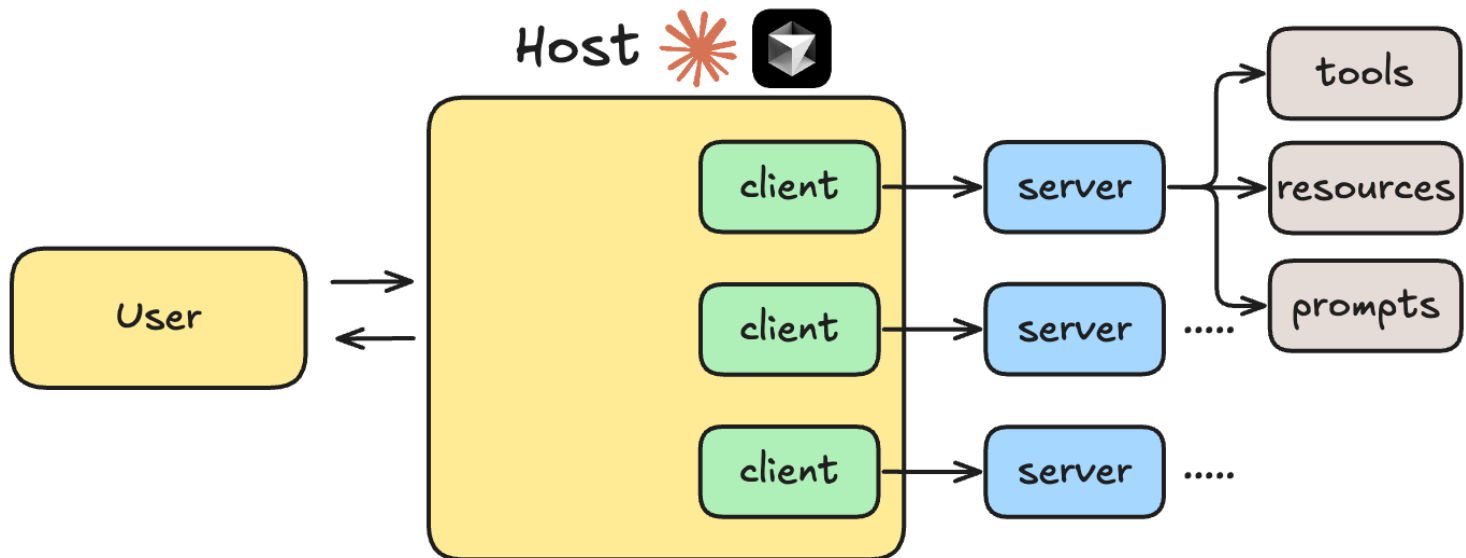


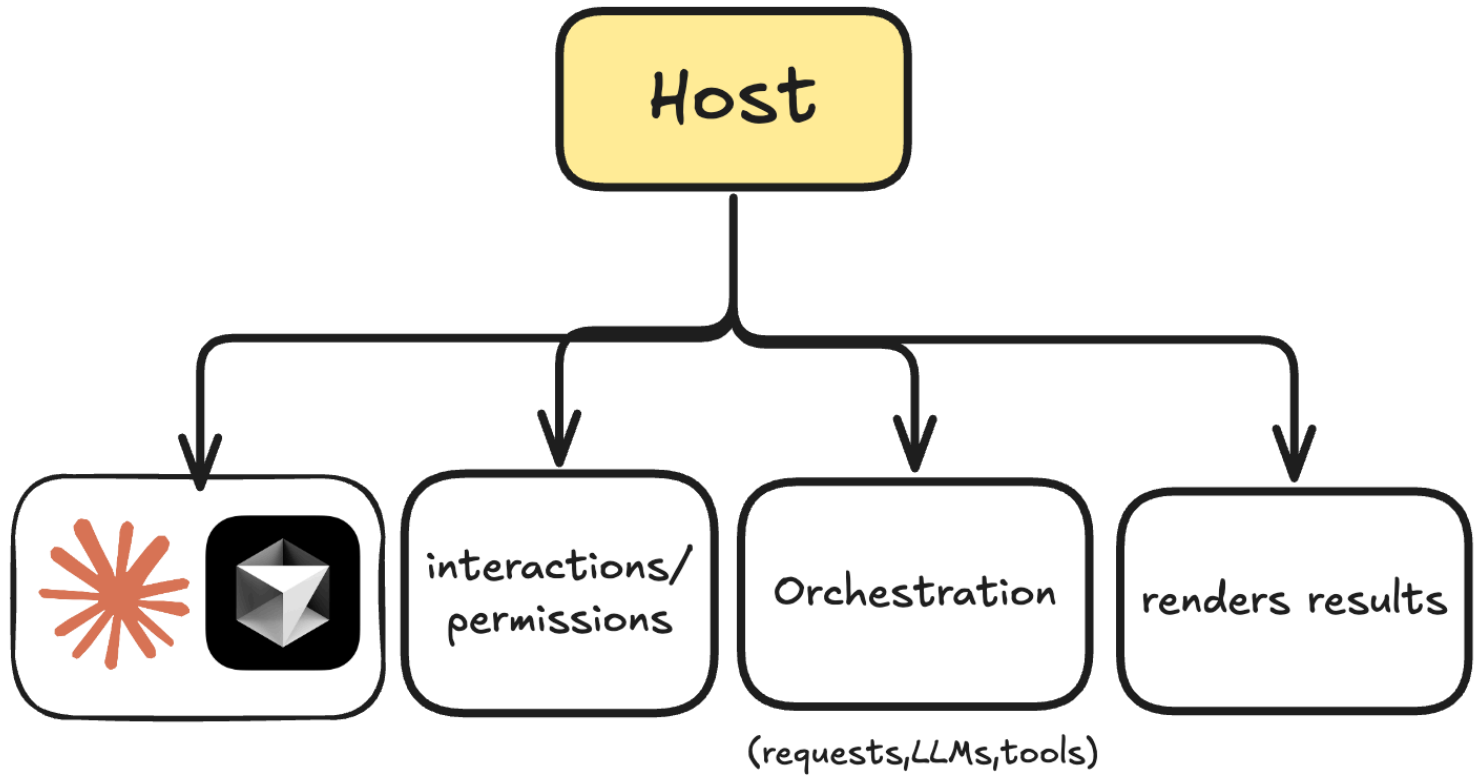


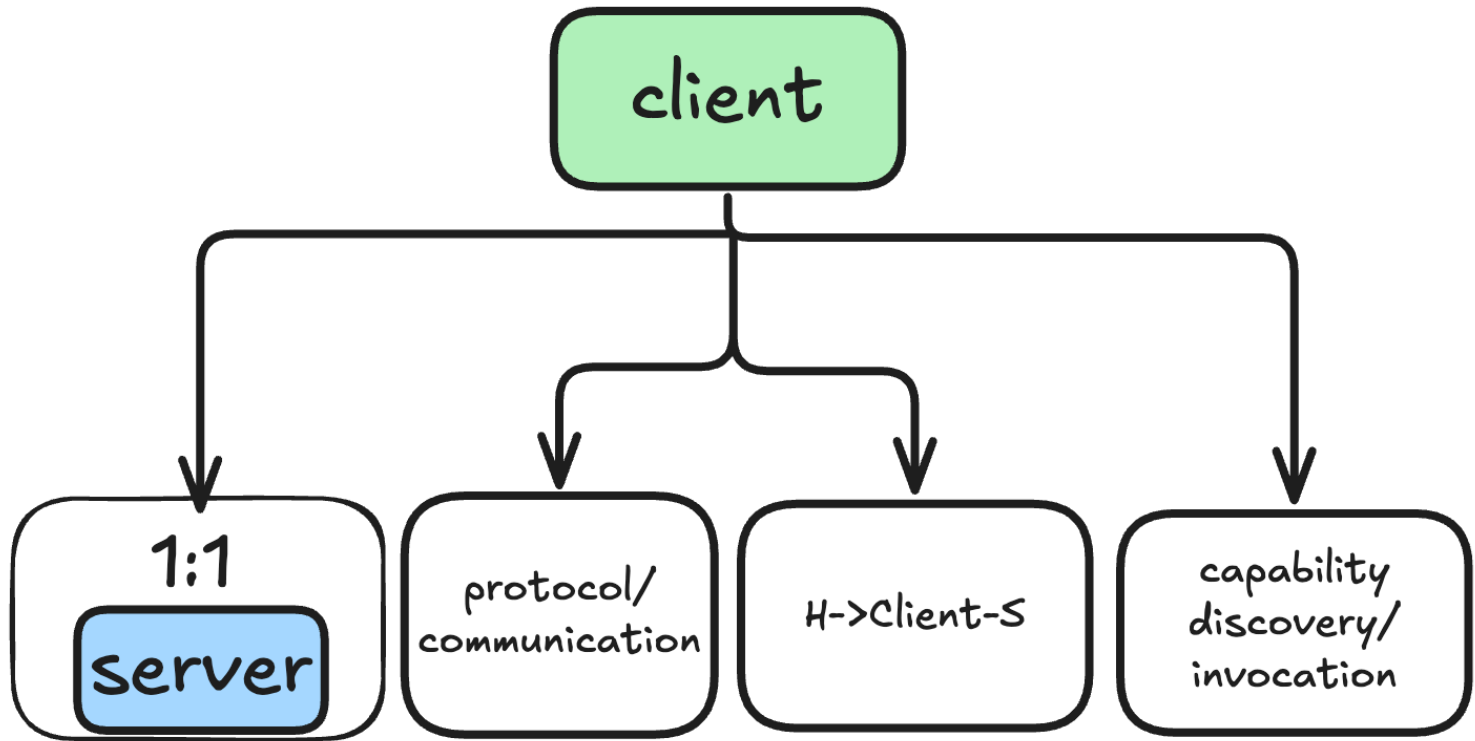


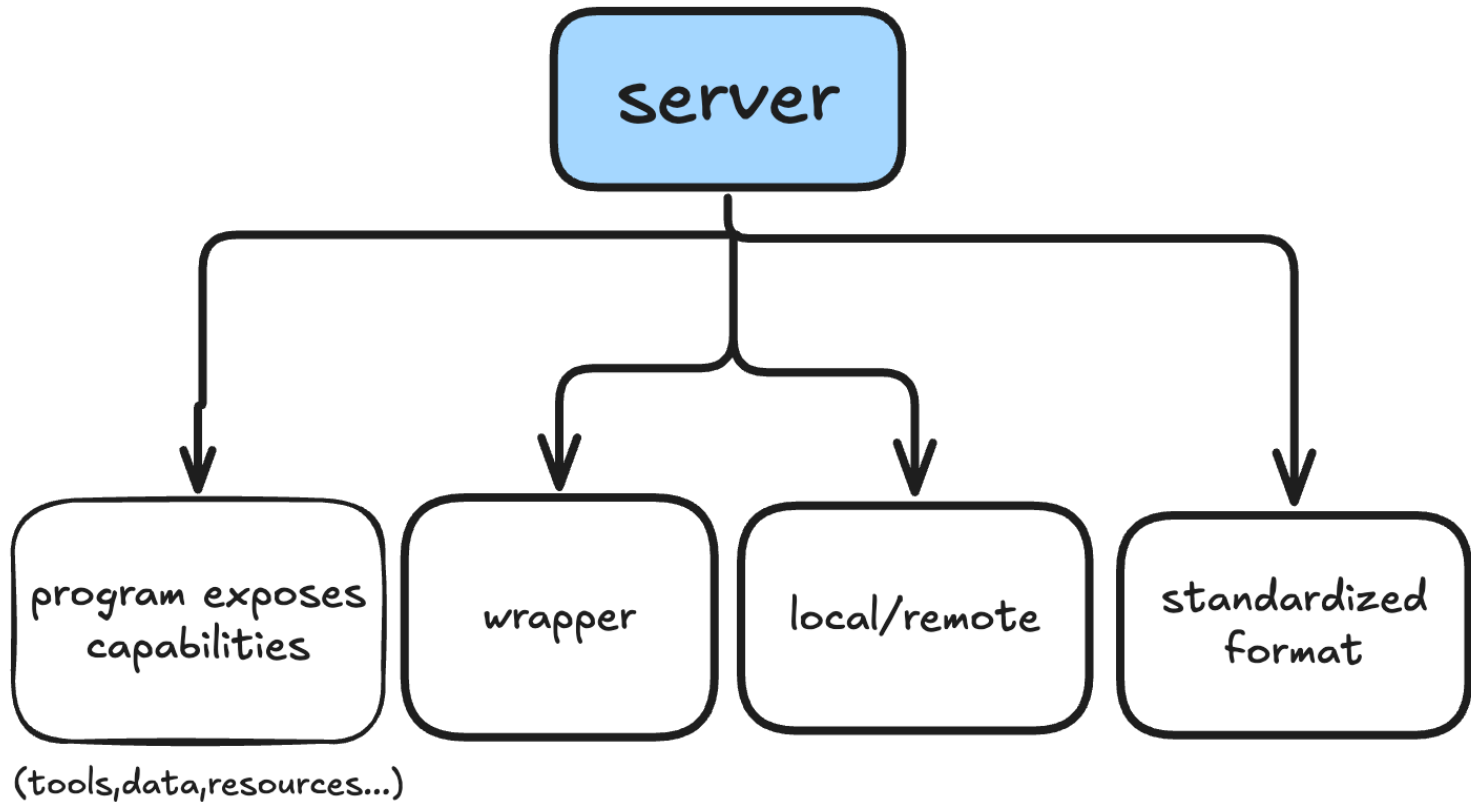




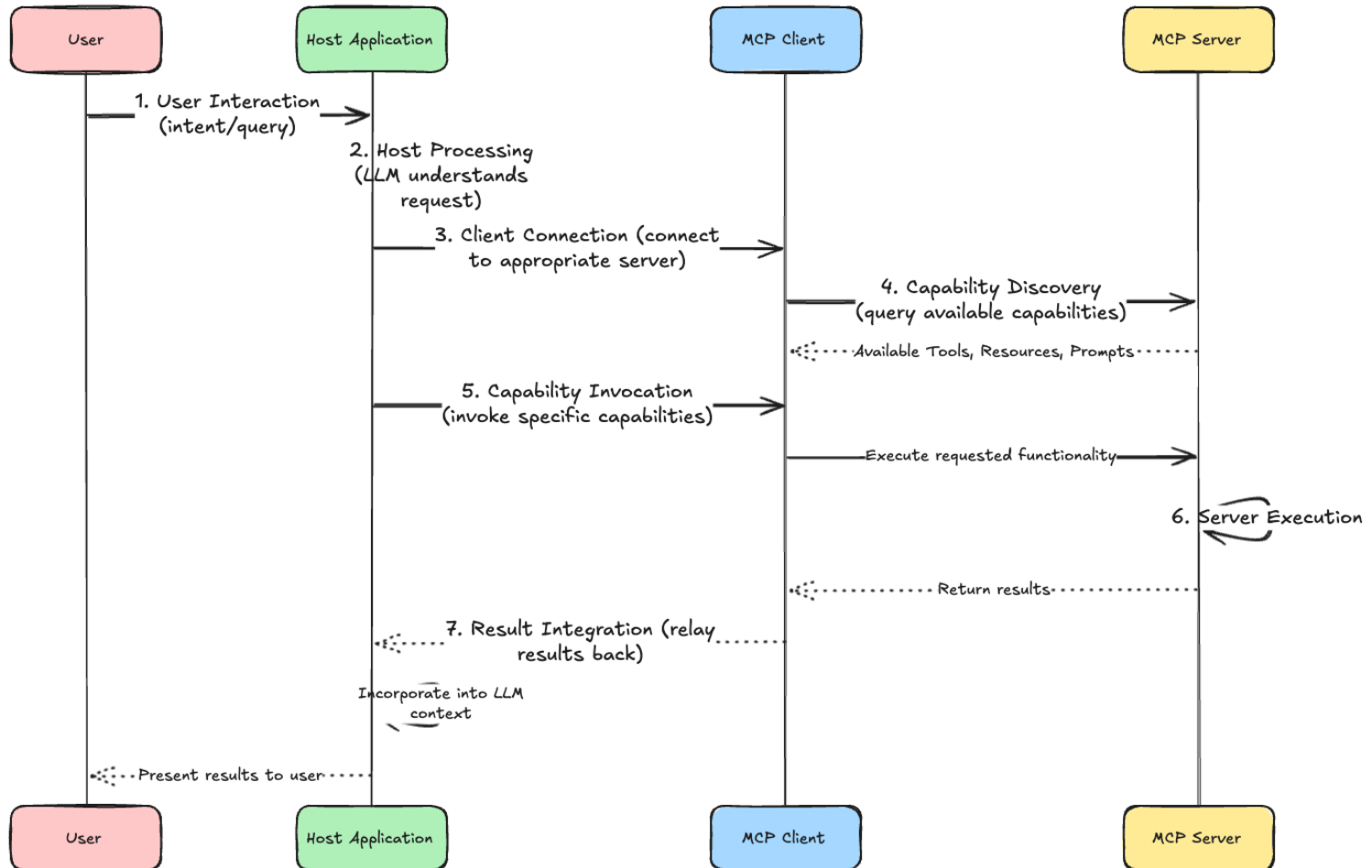








Communication Flow



Communication Flow

- 1. User Interaction**
- 2. Host Processing**
- 3. Client Connection**
- 4. Capability Discovery**
- 5. Capability Invocation**
- 6. Server Execution**
- 7. Result Integration**

Demo - Practical Introduction to MCP SDK

Demo - Creating our First MCP Server

(and using it with Claude Desktop!)

MCP Capabilities: Tools, Resources, Prompts & Sampling

MCP Capabilities

Core Primitives

- **Tools**

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- **Tools**
- **Resources**

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- **Tools**
- **Resources**
- **Prompts**

Core Primitives

- **Tools**
- **Resources**
- **Prompts**
- **Sampling**

Tools

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    Send an email to the given address
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- **Tools** are the most powerful MCP capabilities

Resources

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- Application-controlled data access

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- Application-controlled data access
- Read-only operations

Resources

- Application-controlled data access
- Read-only operations
- Example: File contents, database records

```
def get_file_contents(file_path: str) -> str:  
    """  
    Get the contents of a file  
    """  
    ... # Implementation logic  
    return {  
        "contents": "File contents"  
    }
```

Prompts

- User-controlled templates

Prompts

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- Structure interactions

Prompts

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- Guide workflows

Prompts

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- Example: Code review templates

```
def plan_project(project_name: str) -> str:
    """
    Plan a project
    """
    ... # Implementation logic
    return {
        "plan": "Project plan"
    }
```

Sampling

- Server-initiated LLM interactions

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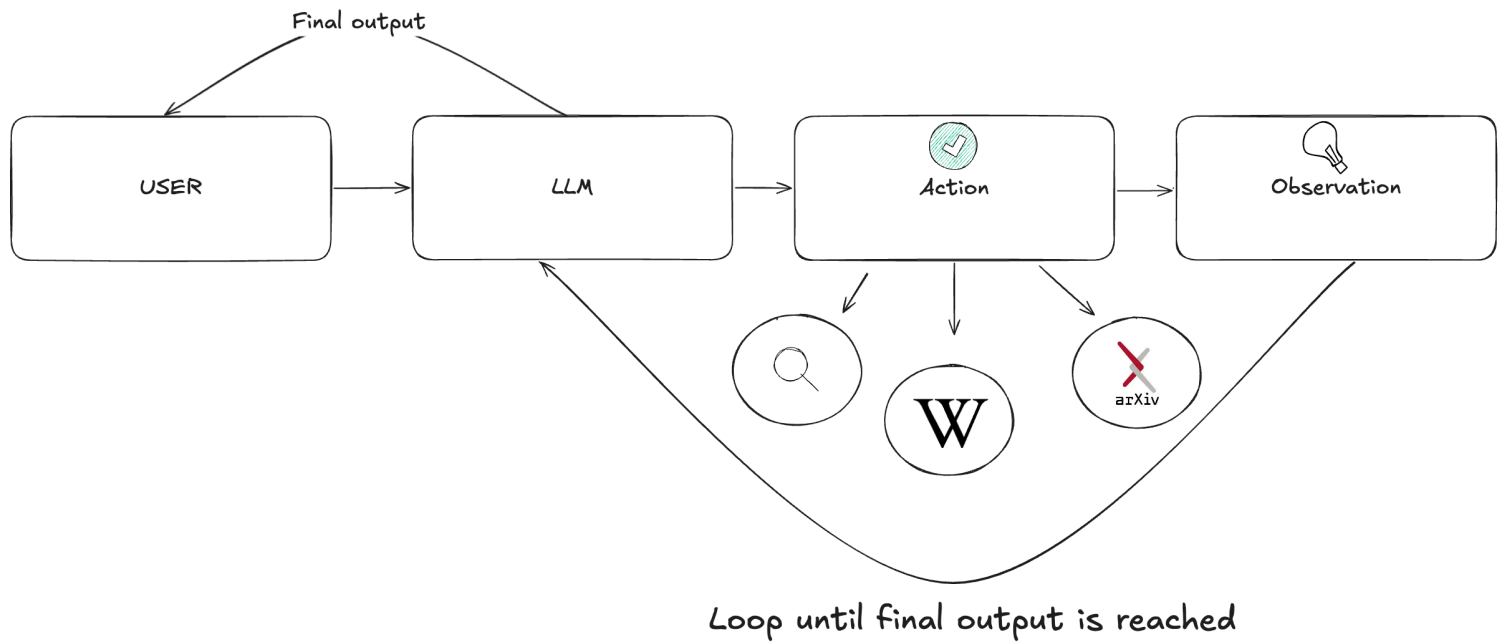
- Server-initiated LLM interactions
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- Example: Multi-step analysis

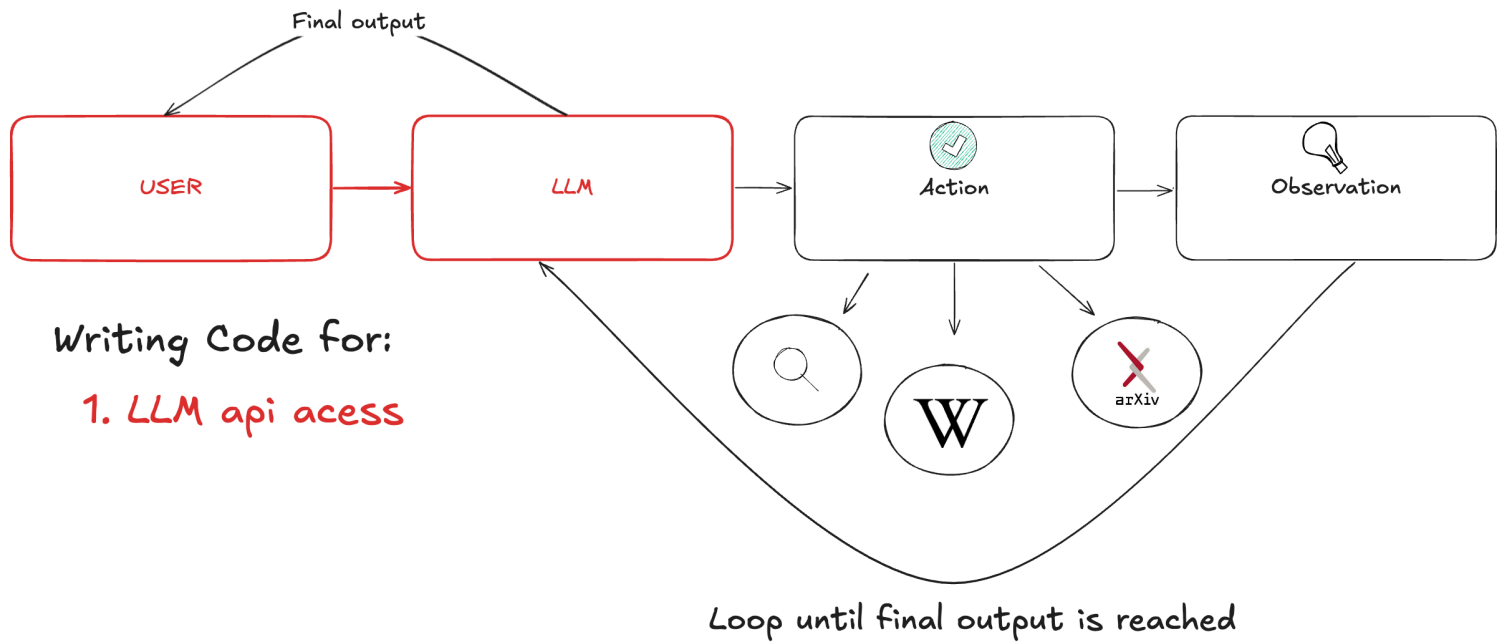
```
def request_sampling(messages, system_prompt=None, include_context="none"):
    """Request LLM sampling from the client."""
    ... # Implementation logic
    return {
        "role": "assistant",
        "content": "Analysis of the provided data..."
    }
```

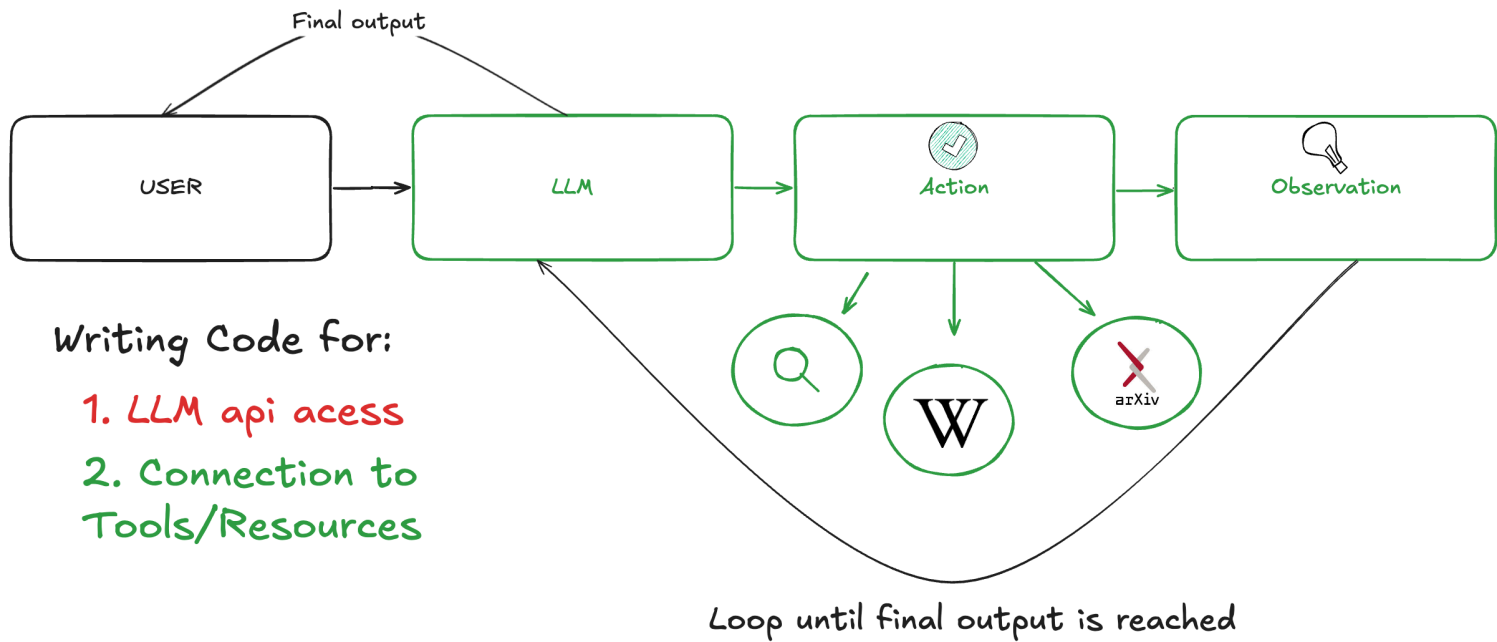
Whiteboard - How MCP Capabilities Work Together

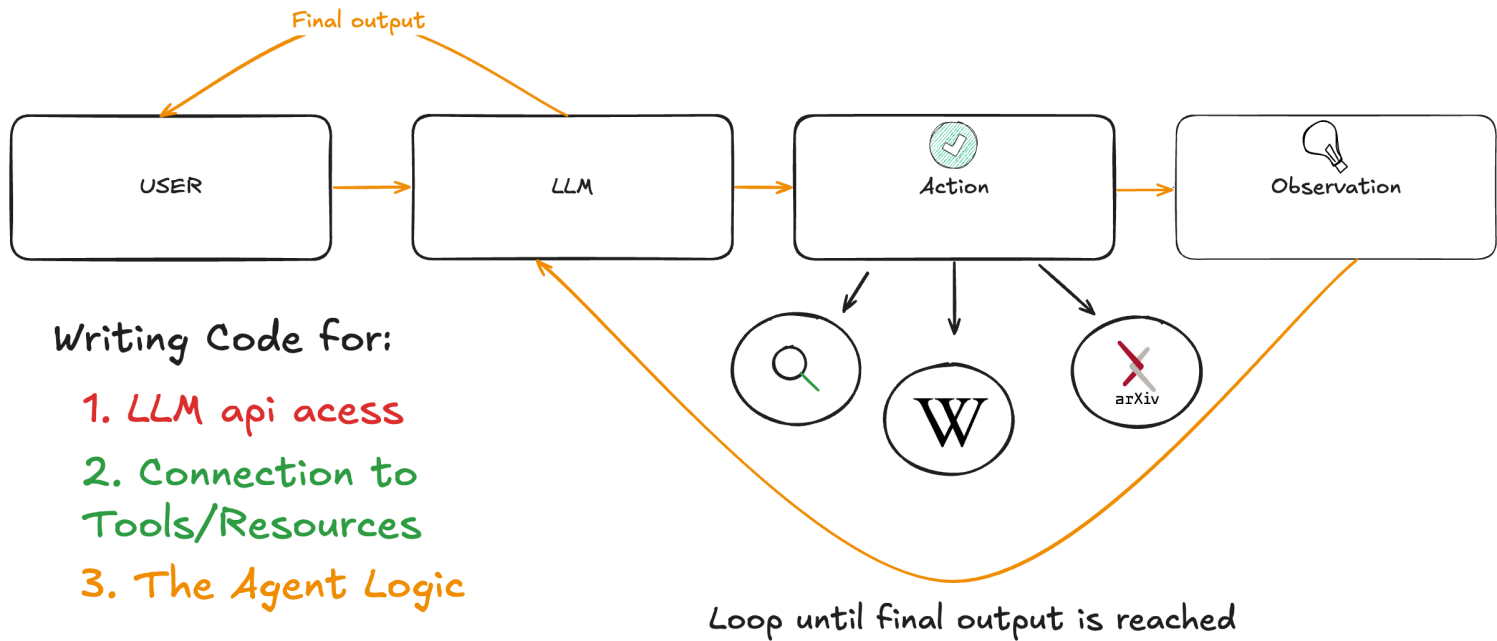
Demo - Implementing MCP Tools, Resources, Prompts & Sampling

Building Agents with MCP

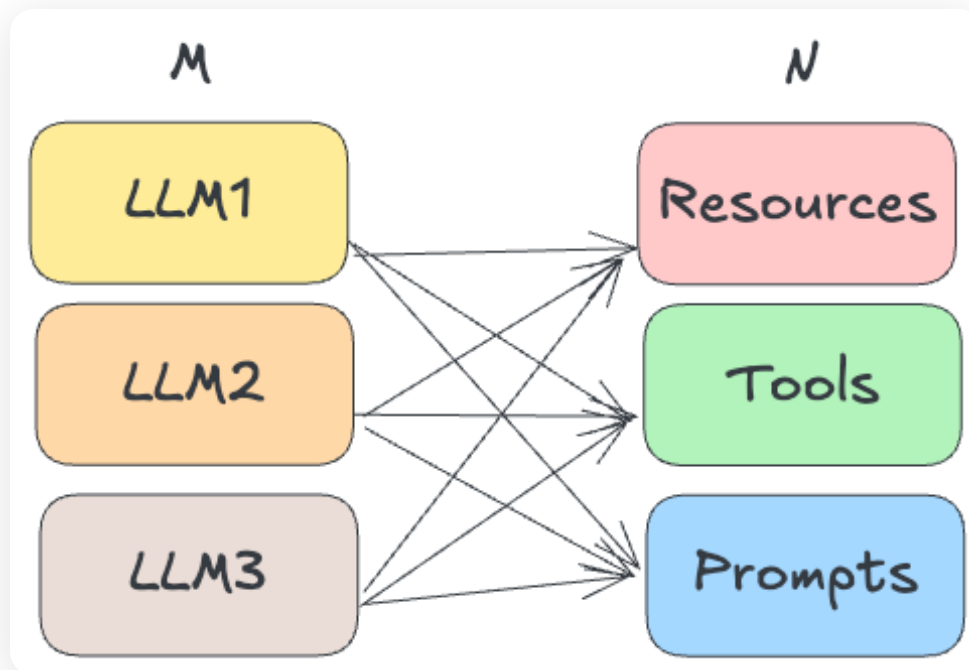




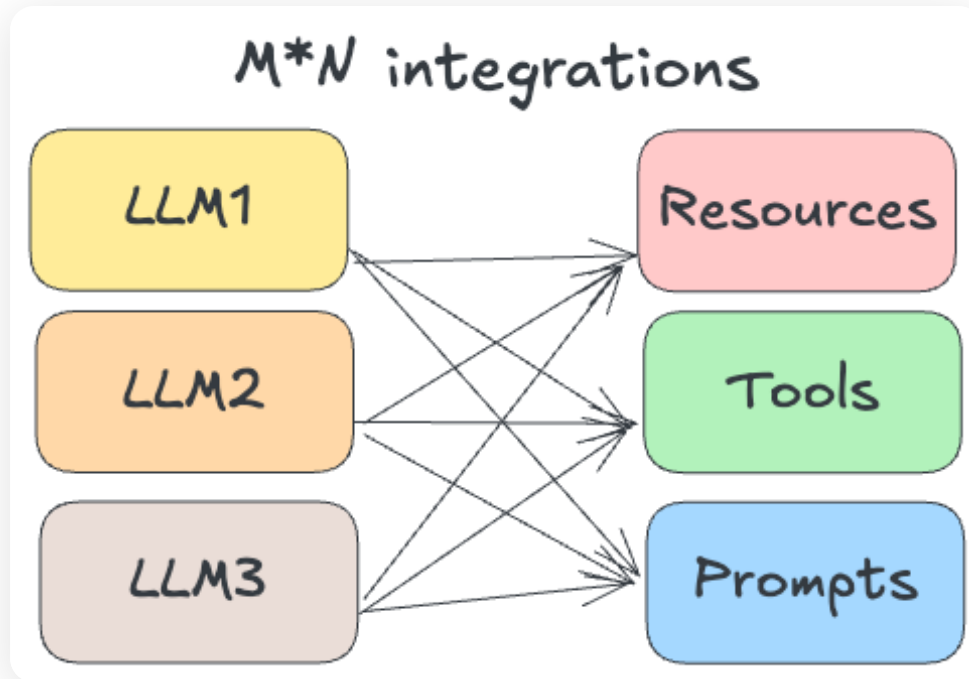




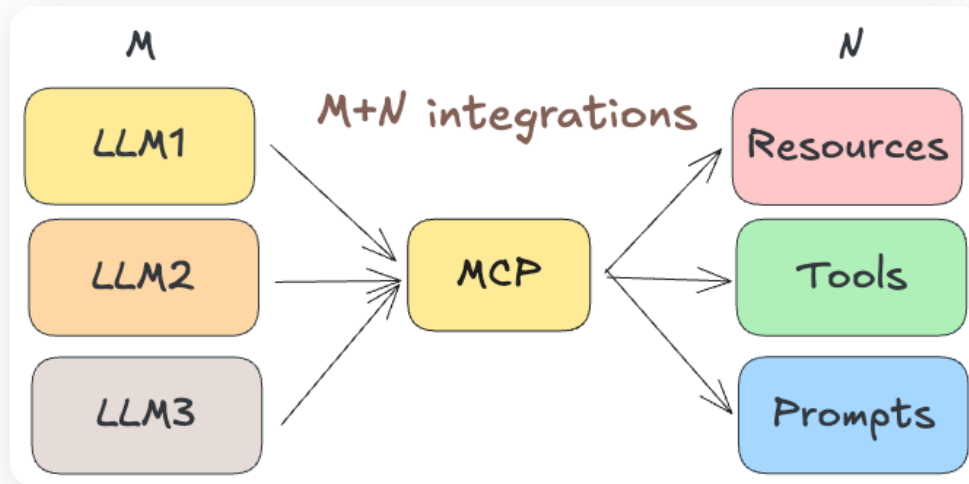
The MN Integration Problem: Multiple LLMs



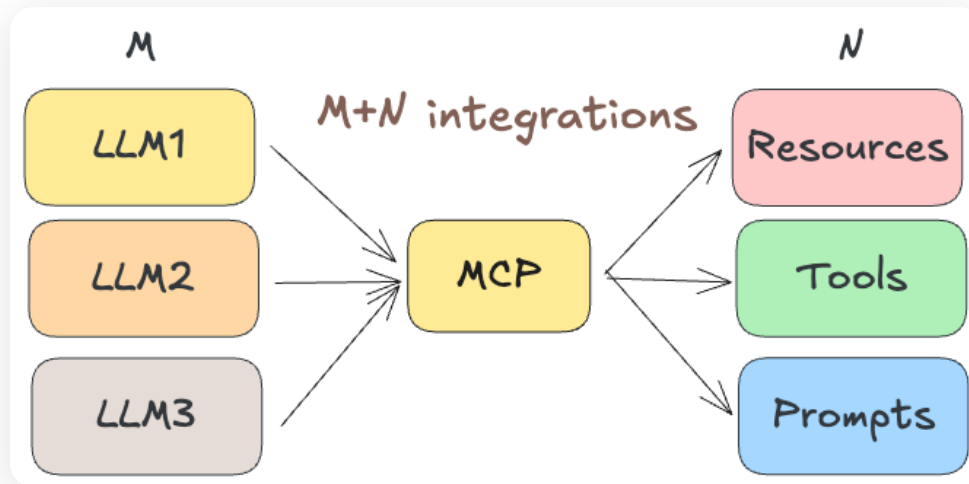
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MCP Simplifies the Integration of Problem for Agent Development



Whiteboard - Agent Development in the Era of MCP

Demo - Building Agents with MCP Using Google's ADK

Demo - Building Agents with MCP Using LangGraph

Demo - Building Agents with MCP Using OpenAI's Agent SDK

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- Multi-Language Support (Python, TypeScript, Swift, Kotlin, Java, Go)

Fun Demo Time! - Using MCP from Claude Desktop and Cursor! Hacks, Tips, and Tricks!

Tool Search: Scaling MCP to Thousands of Tools

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 [Anthropic Docs — Tool Search](#)

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- This doesn't scale — and it puts the burden on developers, not the protocol

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 [Cloudflare Code Mode](#) · [Anthropic Code Execution](#)

The Growing MCP Ecosystem

MCP Market Map

A work in progress.

Top MCP Clients

Chat apps



Coding



Task Automation

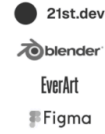


Top MCP Servers

Database



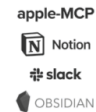
Art & Design



Debugging



Productivity Tools



Email



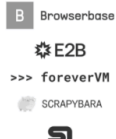
Evaluation



Payments



Agent Execution Environments



Ticketing



Monitoring & Observability



Scraping & Search



MCP Marketplace



Server Generation & Curation



Server Hosting



Connection Management



Charts provided herein are for informational purposes only and should not be relied upon when making any investment decision. Past performance is not indicative of future results. None of the above should be taken as investment advice; please see a16z.com/disclosures for more information.

MCP Security Considerations

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Root cause: No integrity verification between what users approve and what models execute

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All three were demonstrated successfully against Cursor

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- Enforce cross-server dataflow boundaries (server A cannot instruct tools from server B)

Defense Strategies (cont.)

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Key Takeaway: Trust in MCP must be *explicit and verified*, not assumed

MCP Security: Key Takeaways

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 [Invariant Labs — MCP Tool Poisoning Attacks](#)

Whiteboard + Demo - Practical MCP Security Tips

The Protocol "Wars"

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MCP vs A2A vs ACP

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Current Reality: Fragmentation risk vs innovation through competition

The Protocols at a Glance

Protocol	By	Solves
MCP	Anthropic	Agent ↔ Tools & data
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Each fills a different edge in the agent communication graph

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